

# SMART SERVICE APPOINTMENT & WARRANTY AUTOMATION

## Phase 5: Apex Programming (Developer)

**Goal:** Add advanced automation and reusable backend logic to manage Service Appointments and Warranties. This Phase 5 ensures **Smart Service Appointment & Warranty Automation** project is not only declarative (Flows) but also **developer-grade** with Apex for scalability, reusability, and async processing.

### 1. Classes & Objects

- Create a **WarrantyService** class → reusable methods to:
  - Check if a warranty is expired.
  - Update warranty status.
- Create a **ServiceAppointmentService** class → reusable methods to:
  - Validate appointment scheduling.
  - Assign technicians dynamically.

The screenshot displays the Salesforce 'Apex Classes' page. At the top, there's a navigation bar with 'Setup', 'Home', and 'Object Manager'. Below this is a search bar and a sidebar with various setup options. The main content area is titled 'Apex Classes' and includes a green status bar indicating 'Percent of Apex Used: 0.01%'. Below this, there's a table of Apex classes. The table has columns for Action, Name, Namespace Prefix, Api Version, Status, Size Without Comments, and Last Modified By. Two classes are listed: 'ServiceAppointmentService' and 'WarrantyService', both with an API version of 64.0 and an Active status.

| Action                                                                | Name                      | Namespace Prefix | Api Version | Status | Size Without Comments | Last Modified By                   |
|-----------------------------------------------------------------------|---------------------------|------------------|-------------|--------|-----------------------|------------------------------------|
| <a href="#">Edit</a>   <a href="#">Del</a>   <a href="#">Security</a> | ServiceAppointmentService |                  | 64.0        | Active | 43                    | Bhavitha Bolla 9/20/2025, 10:57 PM |
| <a href="#">Edit</a>   <a href="#">Del</a>   <a href="#">Security</a> | WarrantyService           |                  | 64.0        | Active | 634                   | Bhavitha Bolla 9/20/2025, 10:57 PM |

### 2. Apex Triggers

We'll create two triggers:

1. **WarrantyTrigger** → update warranty status when records are created/updated.

**Purpose:**

2. When a warranty is inserted/updated → check if it's expired → set status.

**ServiceAppointmentTrigger** → prevent overlapping bookings + auto-assign technicians.

The screenshot shows the Salesforce Object Manager interface for the 'Service Appointment' object. The top navigation bar includes 'Setup', 'Home', and 'Object Manager'. The breadcrumb trail is 'SETUP > OBJECT MANAGER'. The main heading is 'Service Appointment'. On the left, a sidebar lists various object configuration options: Details, Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, and Restriction Rules. The main content area is titled 'Triggers' and shows '1 Items, Sorted by Label'. A table lists the triggers with columns: LABEL, API VERSION, SIZE WITHOUT COM..., and MODIFIED BY. One trigger is listed: 'serviceAppointment' with API version 64.0 and size 73, modified by 'Bhavitha Bolla' on 9/20/2025 at 11:28 PM. A 'Quick Find' search bar and a 'New' button are located at the top right of the table.

| LABEL              | API VERSION | SIZE WITHOUT COM... | MODIFIED BY                         |
|--------------------|-------------|---------------------|-------------------------------------|
| serviceAppointment | 64.0        | 73                  | Bhavitha Bolla, 9/20/2025, 11:28 PM |

### 3. Trigger Design Pattern

- Instead of writing logic in the trigger, use a **Handler class** (e.g., WarrantyTriggerHandler, ServiceAppointmentTriggerHandler) to keep triggers clean and maintainable.

Q Apex

Email

Apex Exception Email

Custom Code

Apex Classes

Apex Settings

Apex Test Execution

Apex Test History

Apex Triggers

Environments

Jobs

Apex Flex Queue

Apex Jobs

Didn't find what you're looking for?  
Try using Global Search.

SETUP

Apex Triggers

## Apex Triggers

This page allows you to view and modify all the triggers in your organization. To create a new trigger, navigate to the app page.

✓

**Percent of Apex Used: 0.02%**  
You are currently using 1,312 characters of Apex Code (excluding comments and @isTest annotated classes) in your c  
allowed limit of 6,000,000 characters. Note that the amount in use includes both Apex Classes and Triggers defined in

[Compile all triggers](#) ⓘ  
View: All [Create New View](#)

A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U

| Action                                     | Name ↑                             | Namespace Prefix | sObject Type                        | Api Version | Status | Size Without Comments | Last Modified                            |
|--------------------------------------------|------------------------------------|------------------|-------------------------------------|-------------|--------|-----------------------|------------------------------------------|
| <a href="#">Edit</a>   <a href="#">Del</a> | <a href="#">serviceAppointment</a> |                  | <a href="#">Service Appointment</a> | 64.0        | Active | 73                    | <a href="#">Bhavi</a><br>9/20/2017 11:28 |
| <a href="#">Edit</a>   <a href="#">Del</a> | <a href="#">WarrantyTrigger</a>    |                  | <a href="#">Warranty</a>            | 64.0        | Active | 233                   | <a href="#">Bhavi</a><br>9/20/2017 11:18 |

## 4. SOQL & SOSL

- Query for **Available Technicians** where Status\_\_c = 'Available'.

Technician\_\_c@5:30 PM

```
SELECT Id, Name, Phone__c, Email__c FROM Technician__c WHERE Status__c = TRUE
```

Query Results - Total Rows: 0

| Id | Name | Phone__c | Email__c |
|----|------|----------|----------|
|----|------|----------|----------|

Query Grid: Save Rows Insert Row Delete Row Refresh Grid Access in Salesforce: Create New Open Detail Page Edit Page

Logs Tests Checkpoints **Query Editor** View State Progress Problems

```
SELECT Id, Name, Phone__c, Email__c
FROM Technician__c
WHERE Status__c = TRUE
```

History

Executed

SELECT Id, Name, Phone\_\_c, Email\_\_c

Any query errors will appear here...

## 5. Collections (List, Set, Map)

### warrantyTriggerHandler

- ? List → Collects warranties to update in one go.
- ? Set → Ensures no duplicate Customer IDs are collected.
- ? Map → Lets you quickly retrieve a warranty by Id if needed later.

Q apex

▼ Email

Apex Exception Email

▼ Custom Code

Apex Classes

Apex Settings

Apex Test Execution

Apex Test History

Apex Triggers

▼ Environments

▼ Jobs

Apex Flex Queue

Apex Jobs

Didn't find what you're looking for?  
Try using Global Search.

SETUP

Apex Triggers

Apex Trigger

WarrantyTrigger

Help for this Page ?

Apex Trigger Detail

EditDeleteDownloadShow Dependencies

|                  |                                     |                  |                                    |
|------------------|-------------------------------------|------------------|------------------------------------|
| Name             | WarrantyTrigger                     | sObject Type     | Warranty                           |
| Code Coverage    | 0% (0/1)                            | Status           | Active                             |
| Created By       | Bhavitha Bolla, 9/20/2025, 11:18 PM | Last Modified By | Bhavitha Bolla, 9/21/2025, 5:31 AM |
| Namespace Prefix |                                     |                  |                                    |

Apex Trigger

Version SettingsTrace Flags

1trigger WarrantyTrigger on Warranty\_\_c (before insert, before update) {  
2  WarrantyTriggerHandler.handleTrigger(trigger.new, trigger.oldMap);  
3}

EditDeleteDownloadShow Dependencies



Warranty  
**W-00002**

New Contact

Status

Expired,

End Date

9/15/2025

Related

Details

Warranty Name

laptop

Warranty Number

W-00002

Start Date

9/1/2025

End Date

9/15/2025

Status

Expired,

Product

Customer

Expiry Date

Issue Date

Owner



Bhavitha Bolla

Activity

Calendar icon

Dropdown arrow

Checklist icon

Dropdown arrow

Filters: /

Upcoming & Overdue

No activity

Get started by sending an email

No past activity. Past meetings are

## 6. Control Statements

- IF Service Appointment dates overlap for the same Technician → throw error.

Green check mark is the executed one\*\*

The screenshot shows an IDE with three tabs: `Bhavitha_ServiceAppointmentHandler.apex`, `Bhavitha_SA_OverlapTrigger.apex`, and `Bhavitha_SA_OverlapTriggerTest.apex`. The active tab contains the following Apex test code:

```

2 public class Bhavitha_SA_OverlapTriggerTest {
3     @isTest static void testOverlappingAppointments() {
4         // 1 Create a Technician record (custom object) instead of User
5         Technician__c tech = new Technician__c(
6             Name = 'Test Technician'
7         );
8         insert tech;
9
10        // 2 Insert first Service Appointment
11        Service_Appointment__c sal = new Service_Appointment__c(
12            Technician__c = tech.Id,
13            Start_Date__c = Date.today(),
14            End_Date__c = Date.today().addDays(1)
15        );
16        insert sal;
17
18        // 3 Insert overlapping Service Appointment
19        Service_Appointment__c sa2 = new Service_Appointment__c(
20            Technician__c = tech.Id,
21            Start_Date__c = Date.today(),

```

Below the code editor, there is a table showing test results:

| Test Run             | Enqueued Time | Duration | Failures | Total | Overall Code Coverage              |
|----------------------|---------------|----------|----------|-------|------------------------------------|
| TestRun @ 8:45:24 pm |               |          | 1        | 1     | Class                              |
| TestRun @ 8:51:45 pm |               |          | 1        | 1     | Overall                            |
| TestRun @ 9:01:43 pm |               |          | 1        | 1     | Bhavitha_SA_OverlapTrigger         |
| TestRun @ 9:01:57 pm |               |          | 1        | 1     | Bhavitha_ServiceAppointmentHandler |
| TestRun @ 9:06:35 pm |               |          | 1        | 1     | CollectionExamples                 |
| TestRun @ 9:09:09 pm |               |          | 0        | 1     | ServiceAppointmentTriggerHandler   |

Below the table, there is a summary of code coverage:

| Class                              | Percent | Lines |
|------------------------------------|---------|-------|
| Overall                            | 26%     |       |
| Bhavitha_SA_OverlapTrigger         | 100%    | 3/3   |
| Bhavitha_ServiceAppointmentHandler | 100%    | 11/11 |
| CollectionExamples                 | 0%      | 0/2   |
| ServiceAppointmentTriggerHandler   | 0%      | 0/11  |

## 7. Batch Apex

- A scheduled batch job to run nightly:
  - Find all warranties where `End_Date__c < TODAY`.
  - Update status to “Expired.”

The screenshot shows an IDE with a tab for `Bhavitha_ExpiredWarrantyBatch.apex`. Below the code editor, there is a table showing the execution log:

| Timestamp    | Event            | Details                                                                                                                          |
|--------------|------------------|----------------------------------------------------------------------------------------------------------------------------------|
| 21:33:35:001 | USER_INFO        | [EXTERNAL]]005gk000007px0Q[Bhavithabolla2005601@agentforce.com](GMT-07:00) Pacific Daylight Time (America/Los_Angeles) GMT-07:00 |
| 21:33:35:002 | EXECUTION_ST...  |                                                                                                                                  |
| 21:33:35:002 | CODE_UNIT_ST...  | [EXTERNAL]]01pgk000005h90X[Bhavitha_ExpiredWarrantyBatch                                                                         |
| 21:33:35:004 | HEAP_ALLOCATE    | [EXTERNAL]]Bytes:4                                                                                                               |
| 21:33:35:021 | HEAP_ALLOCATE    | [95]]Bytes:3                                                                                                                     |
| 21:33:35:021 | HEAP_ALLOCATE    | [100]]Bytes:152                                                                                                                  |
| 21:33:35:021 | HEAP_ALLOCATE    | [417]]Bytes:408                                                                                                                  |
| 21:33:35:021 | HEAP_ALLOCATE    | [430]]Bytes:408                                                                                                                  |
| 21:33:35:021 | HEAP_ALLOCATE    | [317]]Bytes:6                                                                                                                    |
| 21:33:35:021 | HEAP_ALLOCATE    | [EXTERNAL]]Bytes:11                                                                                                              |
| 21:33:35:021 | SYSTEM_METH...   | [15]]BatchableContextImpl.BatchableContextImpl()                                                                                 |
| 21:33:35:021 | STATEMENT_EX...  | [15]]                                                                                                                            |
| 21:33:35:021 | SYSTEM_METH...   | [15]]BatchableContextImpl                                                                                                        |
| 21:33:35:021 | HEAP_ALLOCATE    | [EXTERNAL]]Bytes:12                                                                                                              |
| 21:33:35:021 | HEAP_ALLOCATE    | [EXTERNAL]]Bytes:4                                                                                                               |
| 21:33:35:021 | VARIABLE_SCO...  | [31]]this[Database.BatchableContextImpl]true false                                                                               |
| 21:33:35:021 | VARIABLE_ASSL... | [31]]this[()]0x7d8e1149                                                                                                          |
| 21:33:35:021 | VARIABLE_SCO...  | [31]]jobId Id false false                                                                                                        |
| 21:33:35:022 | VARIABLE_ASSL... | [31]]jobId 707gk000000kjdQAB                                                                                                     |
| 21:33:35:022 | VARIABLE_SCO...  | [31]]childJobId Id false false                                                                                                   |
| 21:33:35:022 | VARIABLE_ASSL... | [31]]childJobId null                                                                                                             |

Below the log, there is a table showing the execution details:

| User           | Application | Operation  | Time                  | Status  | Read   | Size    |
|----------------|-------------|------------|-----------------------|---------|--------|---------|
| Bhavitha Bolla | Unknown     | Batch Apex | 9/21/2025, 9:33:36 PM | Success | Unread | 3.16 KB |

| W-00002  |           |
|----------|-----------|
| Status   | End Date  |
| Expired, | 9/15/2025 |

| Related         | Details                                                                                                                                                                                             |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Warranty Name   | Owner                                                                                                                                                                                               |
| laptop          |  <a href="#">Bhavitha Bolla</a>  |
| Warranty Number |                                                                                                                                                                                                     |
| W-00002         |                                                                                                                                                                                                     |
| Start Date      |                                                                                                                                                                                                     |
| 9/1/2025        |                                                                                                                                                                                                     |
| End Date        |                                                                                                                                                                                                     |
| 9/15/2025       |                                                                                                                                                                                                     |
| Status          |                                                                                                                                                                                                     |
| Expired,        |                                                                                                                                                                                                     |
| Product         |                                                                                                                                                                                                     |
| Customer        |                                                                                                                                                                                                     |
| Expiry Date     |                                                                                                                                                                                                     |
| Issue Date      |                                                                                                                                                                                                     |
| Claim Amount    |                                                                                                                                                                                                     |

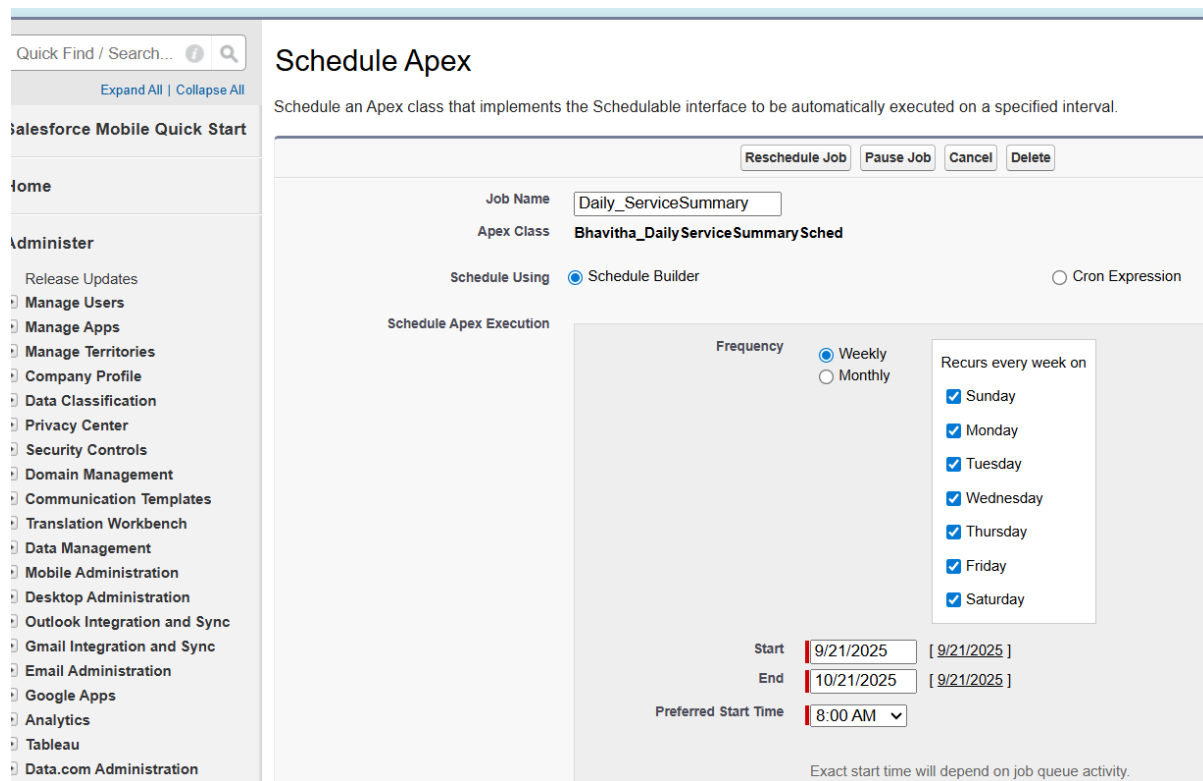
## 8. Queueable Apex

- When a **bulk Service Appointment upload** happens:
  - Queueable job runs async to assign technicians based on availability.
- When **multiple warranties** need recalculation (after policy change), queue them for async processing.

## 9. Scheduled Apex

- Every morning → send email to Service Manager with:

- Today's scheduled Service Appointments.



## 10. Future Methods

- Call an external system (e.g., Manufacturer API) to validate warranty claims asynchronously.
- Example: @future(callout=true) method to check extended warranty eligibility.

## 11. Exception Handling

- Catch overlap errors in appointments and return meaningful error messages to users.

Code Coverage: None API Version: 64 Run Test Go To

```

17 // Insert second overlapping appointment - should fail
18 Service_Appointment__c sa2 = new Service_Appointment__c(
19     Name = 'A2',
20     Technician__c = tech.Id,
21     Start_Date__c = Date.today(),
22     End_Date__c = Date.today().addDays(2)
23 );
24
25 Boolean threw = false;
26 try {
27     insert sa2;
28 } catch (DmlException e) {
29     threw = true;
30     System.assert(e.getMessage().contains('overlapping') || e.
31         'Error message should mention overlapping or suggest a
32     }
33     System.assert(threw, 'Expected overlapping appointment insert
34 }
35 }
36

```

| Status | Test Run | Enqueued Time | Duration | Failures | Total |
|--------|----------|---------------|----------|----------|-------|
| ✓      | Te...    |               | 0        | 1        |       |

| Overall Code Coverage   |            |       |
|-------------------------|------------|-------|
| Class                   | Percent    | Lines |
| <b>Overall</b>          | <b>11%</b> |       |
| Bhavi_SA_OverlapTrigger | 100%       | 2/2   |

### 13. Asynchronous Processing (Batch + Queueable + Future)

- **Batch Apex:** Nightly job → expire warranties.
- **Queueable Apex:** Assign technicians in bulk.
- **Future Method:** External API callouts for warranty validation.